Journal of Mathematical Sciences, Vol. 112, No. 2, 2002

ON ERGODICITY AND STABILITY ESTIMATES FOR SOME NONHOMOGENEOUS MARKOV CHAINS

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UDC 519.2

1. Introduction

Evaluation of the rate of convergence for the ergodic continuous-time Markov chains has been the subject of investigation of the number of authors (see, for example, [1–7]). There is also a growing interest in time-nonhomogeneous Markov chains. Such chains model a variety of queuing systems and they are also used in algorithms of simulated annealing and some other areas. The tool of our study is the method developed in a series of papers by one of the authors. The method is based on two basic ingredients: the logarithmic norm of a semigroup generated by a linear operator and a similarity transformation of the reduced matrix of intensities of the considered Markov chain. In the present paper, we deal with continuous-time countable Markov chains. More accurate estimates give more precise conditions for weak ergodicity and stability for the general-type intensity matrix, stability of birth and death processes, and truncations for Markov chains with bounded jumps. We improve the respective bounds of [9, Theorems 1 and 2; 10, Theorem 4; 11, Theorem 3].

2. General Case

Theorem 1. *Let $X(t)$ be a countable Markov chain with intensity matrix $A(t) = (a_{ij}(t))_{i,j=0}^{\infty}$. Let there exist a sequence $\{d_i\}_{i=1}^{\infty}$ such that $\inf_i d_i = d > 0$ such that*

$$\int^{\infty} \alpha(t)\,dt = +\infty, \tag{1}$$

where

$$\alpha(t) = a(t) - b(t), \quad a(t) = -\sup_k \sum_{i\ge 1} d_i d_k^{-1} a_{ki}(t), \quad b(t) = \sup_k \sum_{i\ge 1} d_i d_k^{-1} a_{0i}(t)$$

are locally integrable on $[0,+\infty)$ functions. Then $X(t)$ is weakly ergodic and for any initial probability distributions $\mathbf{p}^i(s)$, $i = 1, 2$, and any $t \ge s \ge 0$

$$\|\mathbf{p}^1(t) - \mathbf{p}^2(t)\|_{l_1} \le \frac{4}{d} \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big) \sum_{i\ge 1} d_i |\mathbf{p}_i^1 - \mathbf{p}_i^2|. \tag{2}$$

Proof. Write $p_0 = 1 - \sum_{i\ge 1} p_i$ to obtain from the forward Kolmogorov equation the differential equation in the space of sequences l_1

$$\frac{d\mathbf{z}(t)}{dt} = B_1(t)\mathbf{z}(t) + \mathbf{f}(t), \tag{3}$$

where

$$\mathbf{z}(t) = (p_1(t), p_2(t), \dots)^T, \qquad \mathbf{f}(t) = (a_{01}(t), a_{02}(t), \dots)^T, \tag{4}$$

$$B_1(t) = (b_{ij}(t))_{i,j=1}^{\infty}, \qquad b_{ij}(t) = a_{ji}(t) - a_{0i}(t). \tag{5}$$

Let $d = \mathrm{diag}(d_i)$ be a diagonal matrix. Then the space l_{1d} of sequences is defined as

$$l_{1d} = \{\mathbf{z} / \|\mathbf{z}\|_{l_{1d}} = \|d\mathbf{z}\|_{l_1} = \sum_i d_i |z_i| < \infty\}.$$

Proceedings of the Seminar on Stability Problems for Stochastic Models, Eger, Hungary, 2001, Part III.

1072-3374/02/1122-4111$27.00 © 2002 Plenum Publishing Corporation

Let us consider (3) as a differential equation in the space l_{1d}. Then

$$\|B_1(t)\|_{l_{1d}} = \|dB_1(t)d^{-1}\|_{l_1} = \sup_{k\ge1}\sum_{i\ge1} d_i d_k^{-1}|a_{ki}(t) - a_{0i}(t)|$$

$$\le \sup_{k\ge1}\sum_{k\ge1} d_i d_k^{-1}(|a_{ki}(t)| + |a_{0i}(t)|) \le \sup_{k\ge1}\sum_{k\ge1} d_i d_k^{-1}|a_{ki}(t)| + \sup_{k\ge1}\sum_{k\ge1} d_i d_k^{-1} a_{0i}(t)$$

$$\le \sup_{k\ge1}\Big(|a_{kk}(t)| + \sum_{k\ne i} d_i d_k^{-1} a_{ki}(t)\Big) + b(t) \le \sup_{k\ge1}(2|a_{kk}(t)| - a(t)) + b(t) \le \|B(t)\|_{l_1} - a(t) + b(t) = \|B(t)\|_{l_1} - \alpha(t),$$

and both $\mathbf{f}(t)$ and $B_1(t)$ are locally integrable in l_{1d}.

Put $B_1 = B_1^* + B_1^0$, where

$$B_1^*(t) = (b_{ij}^*(t))_{i,j=1}^\infty, \quad b_{ij}^*(t) = a_{ji}(t),$$
$$B_1^0(t) = (b_{ij}^0(t))_{i,j=1}^\infty, \quad b_{ij}^0(t) = -a_{0i}(t).$$

We have for the logarithmic norm of $B_1(t)$ in l_{1d}:

$$\gamma_{l_{1d}}(B_1(t)) \le \gamma_{l_{1d}}(B_1^*(t)) + \gamma_{l_{1d}}(B_1^0(t)) \le \gamma_{l_{1d}}(B_1^*(t)) + \|B_1^0(t)\| \le \gamma_{l_1}(dB_1^*(t)d^{-1}) + \sup_{k\ge1}\sum_{k\ge1} d_i d_k^{-1} a_{0i}(t)$$

$$\le \sup_{k\ge1}\sum_{k\ge1} d_i d_k^{-1} a_{ki}(t) + b(t) \le -a(t) + b(t) \le -\alpha(t). \tag{6}$$

Then the following bound for the Cauchy operator $U(t,s)$ of (3) holds:

$$\|U(t,s)\|_{l_{1d}} \le \exp\Big(\int_s^t \gamma_{l_{1D}}(B_1(\tau))\,d\tau\Big) \le \exp\Big(\int_s^t \alpha(\tau)\,d\tau\Big); \tag{7}$$

hence

$$\|\mathbf{z}_1(t) - \mathbf{z}_2(t)\|_{l_{1d}} \le \|U(t,s)\|_{l_{1d}}\|\mathbf{z}_1(s) - \mathbf{z}_2(s)\|_{l_{1d}} \le \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big)\|\mathbf{z}_1(s) - \mathbf{z}_2(s)\|_{l_{1d}}.$$

This implies (2).

Now let $\overline{X}(t)$ be a perturbed Markov chain with intensity matrix $\overline{A}(t)$ such that

$$\|\hat{A}(t)\|_{l_{1d}} \le (\delta + \omega(t))\alpha(t), \tag{8}$$

where $\hat{A}(t) = \overline{A}(t) - A(t)$, $\delta = \text{const}$, and $\omega(t) \to 0$ as $t \to \infty$.

Theorem 2. *Let the conditions of Theorem* 1 *be satisfied and, in addition, $\alpha(t) \ge 0$ and the sequence $\{d_i\}$ be nondecreasing. Then for any s and t $(0 \le s \le t)$ and for any initial distribution $\mathbf{p} = \mathbf{p}(s)$ the following inequality is fulfilled*:

$$\|\mathbf{p}(P(s,t) - \overline{P}(s,t))\| \le \frac{2}{d_1}\int_s^t \exp\Big(-\int_\tau^t (1 - 2(\delta + \omega(u)))\alpha(u)\,du\Big)\Big[2\exp\Big(-\int_s^\tau \alpha(u)\,du\Big)\|\mathbf{z}(s)\|_{l_{1D}}$$

$$+2d_1\int_s^\tau \exp\Big(-\int_u^\tau \alpha(v)\,dv\Big)b(u)\,du + d_1\Big](\delta + \omega(\tau))\alpha(\tau)\,d\tau, \tag{9}$$

where $P(s,t)$ and $\overline{P}(s,t)$ are transition matrices for $X(t)$ and $\overline{X}(t)$, respectively.

Proof. Write (3) for the perturbed Markov chain

$$\frac{d\overline{\mathbf{z}}(t)}{dt} = \overline{B}_1(t)\overline{\mathbf{z}}(t) + \overline{\mathbf{f}}(t), \tag{10}$$

where $\overline{\mathbf{z}}$, $\overline{B}_1(t)$, and $\overline{\mathbf{f}}$ are the respective characteristics of $\overline{X}(t)$. Rewrite (3) in the following form:

$$\frac{d\mathbf{z}(t)}{dt} = \overline{B}_1(t)\mathbf{z}(t) + \overline{\mathbf{f}}(t) - \hat{B}_1(t)\mathbf{z}(t) - \hat{\mathbf{f}}(t), \tag{11}$$

where $\hat{\mathbf{f}}(t) = \overline{\mathbf{f}}(t) - \mathbf{f}(t)$.

Denote by $\overline{V}(t,s)$ the Cauchy operator for (10). Then we have $\mathbf{z}(s) = \overline{\mathbf{z}}(s)$, and the solution of (10) is given by the well-known formula

$$\overline{\mathbf{z}}(t) = \overline{V}(t,s)\mathbf{z}(s) + \int_s^t \overline{V}(t,\tau)\overline{\mathbf{f}}(\tau)\,d\tau. \tag{12}$$

Now the solution of (11) is

$$\mathbf{z}(t) = \overline{V}(t,s)\mathbf{z}(s) + \int_s^t \overline{V}(t,\tau)\overline{\mathbf{f}}(\tau)\,d\tau - \int_s^t \overline{V}(t,\tau)(\hat{B}_1(\tau)\mathbf{z}(\tau) + \hat{\mathbf{f}}(\tau))\,d\tau, \tag{13}$$

and

$$\mathbf{z}(t) = \overline{\mathbf{z}}(t) - \int_s^t \overline{V}(t,\tau)(\hat{B}_1(\tau)\mathbf{z}(\tau) + \hat{\mathbf{f}}(\tau))\,d\tau. \tag{14}$$

Hence

$$\|\mathbf{z}(t) - \overline{\mathbf{z}}(t)\| \le \int_s^t \|\overline{V}(t,\tau)\|(\|\hat{B}_1(\tau)\|\|\mathbf{z}(\tau)\| + \|\hat{\mathbf{f}}(\tau)\|)\,d\tau. \tag{15}$$

We have the following estimates:

$$\|\hat{B}_1(t)\|_{l_{1d}} = \|d\hat{B}_1(t)d^{-1}\|_{l_1} \le \sup_{k\ge1}\sum_{i\ge1} d_i d_k^{-1}(|\hat{a}_{ki}(t)| + |\hat{a}_{0i}(t)|) \le 2(\delta + \omega(t))\alpha(t), \tag{16}$$

$$\|\hat{\mathbf{f}}(t)\|_{l_{1d}} = \sum_{i\ge1} d_i|\hat{a}_{0i}(t)| \le \sum_{i\ge1} d_{i+1}|\hat{a}_{0i}(t)| \le d_1\|\hat{A}(t)\|_{l_{1d}} \le d_1(\delta + \omega(t))\alpha(t). \tag{17}$$

Using (6), we obtain

$$\gamma_{l_{1d}}(\overline{B}_1(t)) \le \gamma_{l_{1d}}(B_1(t)) + \gamma_{l_{1d}}(\hat{B}_1(t)) \le \gamma_{l_{1d}}(B_1(t)) + \|\hat{B}_1(t)\|_{l_{1d}} \le (2(\delta + \omega(t)) - 1)\alpha(t) \tag{18}$$

and

$$\|\overline{V}(t,s)\|_{l_{1d}} \le \exp\biggl(\int_s^t \gamma_{l_{1d}}(\overline{B}_1(\tau))\,d\tau\biggr) \le \exp\biggl(-\int_s^t (1 - 2(\delta + \omega(\tau)))\alpha(\tau)\,d\tau\biggr). \tag{19}$$

Using (7), we can bound the solution of (11) in the l_{1d}-norm:

$$\|\mathbf{z}(t)\| \le \exp\biggl(-\int_s^t \alpha(\tau)\,d\tau\biggr)\|\mathbf{z}(s)\|_{l_{1d}} + d_1\int_s^t \exp\biggl(-\int_\tau^t \alpha(u)\,du\biggr)b(\tau)\,d\tau. \tag{20}$$

Finally, inequalities (15), (19), and (20) imply the following bound:

$$\begin{aligned}\|\mathbf{z}(t) - \overline{\mathbf{z}}(t)\|_{l_{1d}} \le \int_s^t \exp\biggl(-\int_\tau^t (1 - 2(\delta + \omega(u)))\alpha(u)\,du\biggr)\biggl[2\exp\biggl(-\int_s^\tau \alpha(u)\,du\biggr)\|\mathbf{z}(s)\|_{l_{1D}}\\ +2d_1\int_s^\tau \exp\biggl(-\int_u^\tau \alpha(v)\,dv\biggr)b(u)\,du + d_1\biggr](\delta + \omega(\tau))\alpha(\tau)\,d\tau,\end{aligned} \tag{21}$$

which implies our claim.

3. Stability of Birth and Death Processes

Let D be a triangular matrix,

$$D = \begin{pmatrix} d_0 & d_0 & d_0 & \dots \\ 0 & d_1 & d_1 & \dots \\ 0 & 0 & d_2 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}, \tag{22}$$

where $d_i > 0$, $i = 0, 1, \dots$, and l_{1D} be the respective space of sequences

$$l_{1D} = \{\mathbf{z} / \|\mathbf{z}\|_{l_{1D}} = \|D\mathbf{z}\|_{l_1} < \infty\}.$$

Let $X(t)$ be a birth and death process (BDP) with intensities $a_{i,i+1}(t) = \lambda_i(t)$, $a_{i,i-1}(t) = \mu_i(t)$.

Put

$$\alpha(t) = \inf_{i \ge 0} \left(\lambda_i(t) + \mu_{i+1}(t) - \frac{d_{i+1}}{d_i} \lambda_{i+1}(t) - \frac{d_{i-1}}{d_i} \mu_i(t) \right), \tag{23}$$

where $\{d_i\}$ is a sequence of positive numbers and $d_0 = 1$.

We suppose that for any $0 \le s \le t$

$$\exp\left(- \int_s^t \alpha(\tau)\, d\tau \right) \le N \exp\left(- \int_s^t \beta(\tau)\, d\tau \right), \tag{24}$$

where $N > 0$ and $\beta(t)$ is locally integrable, $0 \le \beta(t) \le \beta$ (see the discussion in [11]).

Now let $\overline{X}(t)$ be a perturbed BDP chain with intensities $\overline{\lambda}_i(t)$ and $\overline{\mu}_i(t)$ such that

$$|\hat{\lambda}_n(t)| \le (\delta + \omega(t))\beta(t), \qquad |\hat{\mu}_n(t)| \le (\delta + \omega(t))\beta(t), \tag{25}$$

where $\hat{\lambda}_n(t) = |\lambda_n(t) - \overline{\lambda}_n(t)|$, $\hat{\mu}_n(t) = |\mu_n(t) - \overline{\mu}_n(t)|$, $\delta = \text{const}$, and $\omega(t) \to 0$ as $t \to \infty$.

Theorem 3. *Let there exist a sequence $\{d_i\}$ such that $d_0 = d_1 = 1$, $\inf d_i = d > 0$, and $\sup_{|i-j|=1} d_i/d_j = m < \infty$, and let (24) be fulfilled. Let for any $t \ge 0$*

$$\lambda_0(t) \le M(t)\beta(t), \tag{26}$$

where $M(t)$ is integrally bounded:

$$\sup_{t \ge 0} \int_t^{t+1} M(\tau)\, d\tau \le M < \infty.$$

Then for any s and t $(0 \le s \le t)$ and for any $\mathbf{p}(s) = \overline{\mathbf{p}}(s)$ the following bound holds:

$$\|\mathbf{p}(t) - \overline{\mathbf{p}}(t)\|_{l_1} \le \frac{4N}{d} \int_s^t \exp\left(- \int_\tau^t [1 - 2(1+m)(\delta + \omega(u))]\beta(u) \right)$$

$$\times \left(2N(1+m) \left(\|\mathbf{z}(s)\|_{l_{1D}} + \beta M \int_\tau^t \exp\left(\int_t^{l+1} \beta(u)\, du \right) dl \right) + 1 \right) (\delta + \omega(\tau))\beta(\tau))\, d\tau, \tag{27}$$

where $\mathbf{z} = (p_1, p_2, \dots)^T$.

Proof. Using (10)–(15), we obtain instead of (16) and (17) the following bounds:

$$\|\hat{B}_1\|_{l_{1D}} = \sup_{i \ge 0} \left(\hat{\lambda}_i(t) + \hat{\mu}_{i+1}(t) + \frac{d_{i+1}}{d_i} \hat{\lambda}_{i+1}(t) + \frac{d_{i-1}}{d_i} \hat{\mu}_i(t) \right)$$

$$\le \sup_{i \ge 0} (\hat{\lambda}_i(t) + \hat{\mu}_{i+1}(t) + m\hat{\lambda}_{i+1}(t) + m\hat{\mu}_i(t)) \le 2(1+m)(\delta + \omega(t))\beta(t), \tag{28}$$

$$\|\hat{\mathbf{f}}(t)\| = d_0 \hat{\lambda}_0(t) \le (\delta + \omega(t))\beta(t). \tag{29}$$

Hence

$$\gamma(\overline{B}_1(t))_{l_{1D}} \le \gamma(B_1(t)) + \|\hat{B}_1(t)\| \le -\alpha(t) + 2(1+m)(\delta+\omega(t))\beta(t). \tag{30}$$

Now we obtain instead of (20) the following inequality:

$$\begin{aligned}
\|\overline{V}(t,s)\|_{l_{1D}} &\le \exp\biggl(\int_s^t \gamma(\overline{B}_1(u))\,du\biggr) \le \exp\biggl(-\int_s^t (\alpha(u) - 2(1+m)(\delta+\omega(u))\beta(u))\,du\biggr) \\
&\le \exp\biggl(-\int_s^t \alpha(u)\,du\biggr)\exp\biggl(-\int_s^t (-2(1+m)(\delta+\omega(u))\beta(u))\,du\biggr) \\
&\le N\exp\biggl(-\int_s^t \beta(u)\,du\biggr)\exp\biggl(-\int_s^t (-2(1+m)(\delta+\omega(u))\beta(u))\,du\biggr) \\
&\le N\exp\biggl(-\int_s^t (1-2(1+m)(\delta+\omega(u)))\beta(u)\,du\biggr).
\end{aligned} \tag{31}$$

We have the following estimate of $\|\mathbf{z}(t)\|$ in the l_{1D}-norm:

$$\begin{aligned}
\|\mathbf{z}(t)\|_{l_{1D}} &\le N\exp\biggl(-\int_s^t \beta(u)\,du\biggr)\|\mathbf{z}(s)\| + \int_s^t N\exp\biggl(-\int_\tau^t \beta(u)\,du\biggr)\|\mathbf{f}(\tau)\|\,d\tau \\
&\le N\exp\biggl(-\int_s^t \beta(u)\,du\biggr)\|\mathbf{z}(s)\| + N\int_s^t \exp\biggl(-\int_\tau^t \beta(u)\,du\biggr)M(\tau)\beta(\tau)\,d\tau.
\end{aligned} \tag{32}$$

Let $t_{i+1} - t_i = 1$, $t_1 = s$, $t_{K-1} < t < t_K$. Then

$$\begin{aligned}
&\int_s^t \exp\biggl(-\int_\tau^t \beta(u)\,du\biggr)M(\tau)\beta(\tau)\,d\tau \le \sum_{i=1}^{K-1}\int_{t_i}^{t_{i+1}} \exp\biggl(-\int_\tau^t \beta(u)\,du\biggr)M(\tau)\beta(\tau)\,d\tau \\
&\quad\le \sum_{i=1}^{K-1}\exp\biggl(-\int_{c_i}^t \beta(u)\,du\biggr)\beta(c_i)\int_{t_i}^{t_{i+1}} M(\tau)\,d\tau \le \sum_{i=1}^{K-1}\exp\biggl(-\int_{c_i}^t \beta(u)\,du\biggr)\beta M \\
&\le \beta M\sum_{i=1}^{K-1}\exp\biggl(-\int_{c_i}^t \beta(u)\,du\biggr) \le \beta M\sum_{i=1}^{K-1}\exp\biggl(-\int_{t_{i+1}}^t \beta(u)\,du\biggr) \le \beta M\int_s^t \exp\biggl(-\int_{\tau+1}^t \beta(u)\,du\biggr)\,d\tau.
\end{aligned} \tag{33}$$

Now (32) and (33) imply the bound

$$\|\mathbf{z}(t)\|_{l_{1D}} \le N\biggl(\|\mathbf{z}(s)\| + \beta M\int_s^t \exp\biggl(\int_t^\tau \beta(u)\,du\biggr)\,d\tau\biggr). \tag{34}$$

Finally, we obtain in the l_{1D}-norm

$$\begin{aligned}
\|\mathbf{z}(t) - \overline{\mathbf{z}}(t)\| &= \int_s^t \|\overline{V}(t,\tau)\|(\|\hat{B}_1(\tau)\|\|\mathbf{z}(\tau)\| + \|\hat{f}(\tau)\|)\,d\tau \\
&\le \int_s^t N\exp\biggl(-\int_\tau^t [1-2(1+m)(\delta+\omega(u))]\beta(u)\,du\biggr) \\
&\times\biggl\{2N(1+m)(\delta+\omega(\tau))\beta(\tau)\biggl(\|\mathbf{z}(s)\| + \beta M\int_\tau^t \exp\biggl(\int_t^{l+1}\beta(u)\,du\biggr)\,dl\biggr) + (\delta+\omega(\tau))\beta(\tau)\biggr\}\,d\tau
\end{aligned}$$

$$\leq \int_s^t N \exp\biggl(- \int_\tau^t [1 - 2(1+m)(\delta + \omega(u))]\beta(u)\, du \biggr)$$

$$\times \biggl[\biggl\{ 2N(1+m) \biggl(\|\mathbf{z}(s)\| + \beta M \int_\tau^t \exp\biggl(\int_t^{l+1} \beta(u)\, du \biggr)\, dl \biggr) + 1 \biggr\} (\delta + \omega(\tau))\beta(\tau) \biggr]\, d\tau. \tag{35}$$

This implies (27).

4. Truncations of Markov Chains with Bounded Jumps

Definition 1. A Markov chain $X(t)$ is called a chain with bounded jumps if there exists N such that $a_{ij}(t) \equiv 0$ if $|i - j| > N$. Let $X_n(t)$ be a "truncated" Markov chain with intensity matrix $A_n(t) = (a_{ij_n}(t))_{i,j=0}^n$, where

$$a_{ij_n}(t) = \begin{cases} a_{ij}(t), & j \neq i,\ j \neq n, \\ -\sum_{k \neq i} a_{ik}(t), & j = i,\ j \neq n, \\ \sum_{k \geq n} a_{ik}(t), & j = n. \end{cases} \tag{36}$$

Theorem 4. *Let $X(t)$ be a nonhomogenious Markov chain with bounded jumps and let $X_n(t)$ be a truncated Markov chain. Let there exist a nondecreasing sequence $\{d_i\}$ such that $d_1 = 1$;*

$$\int^\infty \alpha(t)\, dt = +\infty, \tag{37}$$

where

$$\alpha(t) = a(t) + d_N a_{00}(t), \quad a(t) = -\sup_k \sum_{i \geq 1} d_i d_k^{-1} a_{ki}(t)$$

are locally integrable on $[0, +\infty)$ functions, and, in addition, let there exist $Q < \infty$ such that for almost all $t > 0$

$$|a_{00}(t)| < Q a(t). \tag{38}$$

Then for any $0 \leq s \leq ts$ the following inequality holds:

$$\|\mathbf{p}(t) - \mathbf{p}_n(t)\| \leq \frac{1}{d_{n-N+1}} \int_s^t M(\tau) \biggl(\exp\biggl(- \int_s^\tau \alpha(u)\, du \biggr) \|\mathbf{z}(s)\|_{l_1 D} + d_N Q \biggr)\, d\tau, \tag{39}$$

where $M(t) = \operatorname{ess\,sup}_{0 \leq \tau \leq t} \|A(\tau)\|$ and $\mathbf{p}_n(t)$ is the vector of state probabilities for the truncated Markov chain.

Proof. We shall use the same notions for the finite and infinite vectors and matrices if they have equal nonzero coordinates. Then the forward Kolmogorov system for $X(t)$ can be written in the form

$$\frac{d\mathbf{x}(t)}{dt} = B_n(t)\mathbf{x}(t) + (B(t) - B_n(t))\mathbf{x}(t), \tag{40}$$

where $\mathbf{x}(t) = \mathbf{p}(t)^T$. Let $U_n(t, s)$ be a Cauchy operator for the equation

$$\frac{d\mathbf{x}_n(t)}{dt} = B_n(t)\mathbf{x}_n(t). \tag{41}$$

Let $\mathbf{x}_n(s) = \mathbf{x}(s)$. Then, using (40), we have

$$\mathbf{x}(t) = U_n(t, s)\mathbf{x}_n(s) + \int_s^t U_n(t, \tau)\{(B(\tau) - B_n(\tau))\mathbf{x}(\tau)\}\, d\tau \tag{42}$$

and

$$\mathbf{x}(t) - \mathbf{x}_n(t) = \int_s^t U_n(t,\tau)\{(B(\tau) - B_n(\tau))\mathbf{x}(\tau)\}\,d\tau. \tag{43}$$

Now, applying the known inequality $\|U_n(t,s)\| \le 1$ (for any s and t, $0 \le s \le t$), we obtain

$$\|\mathbf{x}(t) - \mathbf{x}_n(t)\|_{l_1} \le \int_s^t \|U_n(t,\tau)\|_{l_1}\|\{(B(\tau) - B_n(\tau))\mathbf{x}(\tau)\}\|_{l_1}\,d\tau \le \int_s^t \|\{(B(\tau) - B_n(\tau))\mathbf{x}(\tau)\}\|_{l_1}\,d\tau. \tag{44}$$

All elements of the first $n - N$ rows of $B(t) - B_n(t)$ are null for any t. Therefore,

$$\|(B(\tau) - B_n(\tau))\mathbf{x}(\tau)\|_{l_1} \le \|A(\tau)\|_{l_1} \sum_{i=n-N+1}^{\infty} |p_i(\tau)| \le M(\tau) \sum_{i=n-N+1}^{\infty} |p_i(\tau)|. \tag{45}$$

For bounding of $\sum_{i=n-N+1}^{\infty} |p_i(t)|$ we consider (3).
Let us estimate the norm of $\mathbf{z}(t)$ in l_{1D}:

$$\|\mathbf{z}(t)\|_{l_{1D}} \le \|V(t,s)\|_{l_{1D}}\|\mathbf{z}(s)\|_{l_{1D}} + \int_s^t \|V(t,\tau)\|_{l_{1D}}\|\mathbf{f}(\tau)\|_{l_{1D}}\,d\tau. \tag{46}$$

Instead of (6) and (7) we have

$$\gamma(B_1(t))_{l_{1D}} \le \gamma(B_1^*(t))_{l_{1D}} + \gamma(B_1^0(t))_{l_{1D}} \le \gamma(B_1^*(t))_{l_{1D}} + \|B_1(t)\|_{l_{1D}} = \gamma(DB_1^*(t)D^{-1})_{l_1} + \sup_{k\ge1}\sum_{i\ge1} d_i d_k^{-1} a_{0i}(t)$$

$$\le \sup_{k\ge1}\sum_{i\ge1} d_i d_k^{-1} a_{ki}(t) + d^{-1}\max_{i\le N} d_i \sum_{i=1}^{N} a_{0i}(t) \le -a(t) - d_N a_{00}(t) \le -\alpha(t) \tag{47}$$

and

$$\|V(t,\tau)\|_{l_{1D}} \le \exp\Big(\int_\tau^t \gamma(B_1(u))_{l_{1D}}\,du\Big) \le \exp\Big(-\int_\tau^t \alpha(u)\,du\Big). \tag{48}$$

We have

$$\|\mathbf{f}(t)\|_{l_{1D}} = \sum_{i=1}^{N} d_i|a_{i0}(t)| \le \max_{i\le N} d_i \sum_{i=1}^{N} a_{i0}(t) \tag{49}$$

$$= d_N|a_{00}(t)| \le d_N Q\alpha(t). \tag{50}$$

Then we obtain the following bound:

$$\|\mathbf{z}(t)\|_{l_{1D}} \le \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big)\|\mathbf{z}(s)\|_{l_{1D}} + \int_s^t \exp\Big(-\int_\tau^t \alpha(u)\,du\Big)d_N Q\alpha(\tau)\,d\tau$$

$$\le \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big)\|\mathbf{z}(s)\|_{l_{1D}} + \int_s^t \exp\Big(-\int_s^t \alpha(u)\,du + \int_s^\tau \alpha(u)\,du\Big)d_N Q\alpha(\tau)\,d\tau$$

$$\le \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big)\Big\{\|\mathbf{z}(s)\|_{l_{1D}} + d_N Q\int_s^t \exp\Big(\int_s^\tau \alpha(u)\,du\Big)\alpha(\tau)\,d\tau\Big\}$$

$$\le \exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big)\Big\{\|\mathbf{z}(s)\|_{l_{1D}} + d_N Q\exp\Big(\int_s^t \alpha(\tau)\,d\tau\Big)\Big\} \le \|\mathbf{z}(s)\|_{l_{1D}}\exp\Big(-\int_s^t \alpha(\tau)\,d\tau\Big) + d_N Q. \tag{51}$$

On the other hand,

$$d_{n-N+1}\sum_{i=n-N+1}^{\infty}|p_i(\tau)|\le\sum_{i=n-N+1}^{\infty}d_i|p_i(\tau)|\le\|\mathbf{z}(\tau)\|_{l_{1D}},\tag{52}$$

hence

$$\sum_{i=n-N+1}^{\infty}|p_i(\tau)|\le\frac{1}{d_{n-N+1}}\|\mathbf{z}(\tau)\|_{l_{1D}}.\tag{53}$$

Now our assertion follows from inequalities (44), (45), and (53).

Corollary 1. *Let $d_i=d^i$. Then*

$$\|\mathbf{p}(t)-\mathbf{p}_n(t)\|_{l_1}\le d^{N-n-1}\int_s^t M(\tau)\biggl(\exp\biggl(-\int_s^\tau\alpha(u)\,du\biggr)\|\mathbf{z}(s)\|_{l_{1D}}+d^NQ\biggr)\,d\tau.$$

Corollary 2. *Let $\alpha(t)\ge0$ for any $t\ge0$. Then*

$$\|\mathbf{p}(t)-\mathbf{p}_n(t)\|_{l_1}\le\frac{1}{d_{n-N+1}}(\|\mathbf{z}(s)\|_{l_{1D}}+d_NQ)\int_s^t M(\tau)\,d\tau.$$

Corollary 3. *Let the initial probability distribution $\mathbf{p}(s)=\mathbf{e}_0$; then*

$$\|\mathbf{p}(t)-\mathbf{p}_n(t)\|_{l_1}\le\frac{1}{d_{n-N+1}}d_NQ\int_s^t M(\tau)\,d\tau.$$

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